AL-LEN

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I. INTRODUCTION

For many learners, the journey toward quality education begins with foundational elements often taken for granted: reliable access to electricity, modern digital technology, and continuous information streams. Sustainable Development Goal (SDG) 4 promotes inclusive and equitable quality education and lifelong learning opportunities for all; however, realizing this goal remains a significant challenge in geographically isolated regions. Although digital technologies have democratized access to educational resources globally, many communities remain completely beyond the reach of the physical infrastructure required to support digital learning [1]. The educational deprivation in these off-the-grid societies is not a result of a lack of motivation to study, but rather a systemic lack of the necessary infrastructure to access and utilize modern educational technologies.

In the Philippines, this infrastructural divide presents a critical barrier. The Department of Energy defines off-grid communities as areas entirely unreachable by conventional power grids, requiring alternative localized solutions. As of September 2026, the Department reported that there are 369 unserved and underserved regions spanning 23 provinces, leaving a total of 101,468 homes unlit [3]. This lack of electricity is severely compounded by widespread inadequacies in internet connectivity and the unaffordability of electronic devices. According to the Philippine Statistics Authority, while 48.8% of households nationally reported having internet access at home in 2024, the numbers sharply decline in rural regions. Specifically, the Zamboanga Peninsula and the Bangsamoro Autonomous Region in Muslim Mindanao (BARMM) recorded household internet access rates of only 21.2% and 27.7%, respectively [4]. For households without connectivity, the prohibitive costs of internet subscriptions (58.3%) and electronic equipment (37.5%) were cited as the primary obstacles. Consequently, marginalized students in these areas—such as those attending Cagmanaba Elementary School in Northern Samar, which lacks both electricity and communication facilities—are fundamentally excluded from the digital education ecosystem [2].

The technological constraint of conventional digital education solutions is their strict reliance on constant connectivity and a stable alternating current (AC) power supply. The majority of modern online courses and cloud-based learning management systems operate under the assumption that users can ping distant servers in real-time whenever they require educational materials. In communities where the power supply is either non-existent or highly irregular, and where cellular connectivity is severely restricted, this assumption completely

collapses. Merely attempting to improve cellular internet access is insufficient, as households still lack the affordable personal equipment and the continuous power required to utilize it. Therefore, an alternative, decentralized infrastructure must be engineered.

Offline educational technology addresses this gap by facilitating the localized storage of high-density educational materials, allowing for continuous use without requiring an active ISP connection. To solve the persistent connectivity problem, this paper proposes the Delay-Tolerant Network (DTN) Node. The DTN Node is a low-power, solar-driven edge server engineered specifically to deliver interactive digital learning materials to off-grid communities. Instead of requiring consistent internet access, the system utilizes the DTN architectural concept, wherein data is held locally and temporarily until a physical opportunity for communication—such as a mobile data mule—becomes available [1]. This asynchronous routing principle is uniquely suited for environments where end-to-end network paths cannot be guaranteed.

The primary objective of the DTN Node is to aggressively push educational resources directly into the community, shifting the technological burden away from impoverished households. By utilizing a highly portable hardware footprint, a localized multimedia database, off-grid energy harvesting, and tangible STEM learning kits, the DTN Node acts as a permanent, offline digital library. By facilitating better access to educational resources in these vulnerable sectors, the DTN Node seeks to remove infrastructural obstacles and ensure that off-grid students can participate in the modern digital world.

The specific contributions of this paper are summarized below:

1. Creation of a low energy consumption Delay-Tolerant Networking (DTN)-based learning node that can store and distribute multimedia educational material within off-grid communities.
2. Offline provision of educational resources to ensure constant access to multimedia educational content regardless of permanent Internet connections.
3. Integration of a flexible power supply system that allows the proposed learning node to function in off-grid communities heavily restricted by traditional electricity limits.
4. Tangible kits integrated with digital educational resources to serve as additional, physical learning material in cases of insufficient infrastructure.
5. Application of Delay-Tolerant Networking methodology to educational resource distribution as a viable alternative to Internet-based learning systems.

6. An infrastructure-centered approach that targets eliminating the specific barriers to education caused by the lack of electricity, Internet access, and household computer availability.

II. PROPOSED SOLUTION

A. Concept

The proposed system will be based on asynchronous delay tolerant network architecture that deploys permanent digital educational infrastructure in off-grid communities. The two-tier system will comprise Local Edge Nodes and a Data Mule. The former will be a solar-powered Raspberry Pi Zero 2W micro-server that is permanently stationed at the village and utilizes deep-sleep capable 18650 lithium-ion power batteries to permanently host an offline Wi-Fi access point that serves localized progressive web application encyclopedia and computer engineering modules directly into the teachers device which they can then disseminate to the students. Since the permanent node will remain offline, the data mule will offer a mobile gateway to fresh information and educational content from the outside world. Whenever the data mule is present in the village, it will perform a secure proximity handshake with the Local Edge Node to synchronize data uploaded to the cloud in both directions. Downloading updates to the local SQLite database while uploading student performance for analysis in the urban region for professors to create a customized curriculum for what the students need.

B. Hardware

1. The Local Edge Node

The Local Edge Node (LEN) can stand as a permanent, standalone, solar-powered micro-server deployed in off-grid Geographically Isolated and Disadvantaged Areas (GIDA). Its main purpose is to host the Local Progressive Web (PWA) encyclopedia, manage the SQLite database, execute local routing via an offline Software Access Point, and handle asynchronous Delay-Tolerant Networking handshakes with traveling Data Mules.

2. Microcontroller

For the brain of the entire operation, we chose the **Raspberry Pi zero 2W**. It has a 1GHz quad-core 64-bit ARM Cortex-A53 CPU, 512MB LPDDR2 SDRAM, integrated 2.4GHz 802.11 b/g/n Wi-Fi and BLE 4.2). We specifically chose this microcontroller for the following reasons:

Table 1. Core Computing Unit Engineering comparison Matrix

Specification / Feature	**Raspberry Pi Zero 2 W**	**ESP32-WROOM-32**	**Raspberry Pi 4 (4GB)**
Operating System	Full Linux (Raspberry Pi OS)	None (RTOS / Bare Metal)	Full Linux (Raspberry Pi OS)
Software Stack Capability	Nginx, NodeJS, SQLite, PWA	Microdot, Flat JSON Files	Nginx, NodeJS, SQLite, PWA
Idle Power Draw (at 5V)	~0.40 W (80 mA)	~0.02 W (5 mA)	~3.00 W (600 mA)
Full Load Power Draw	~1.75 W (350 mA)	~0.80 W (160 mA)	~6.00 W (1200 mA)
Off-Grid Viability	**High** (Low power + full stack)	High (Limited software cap.)	Low (Rapid battery drain)

As you can see in the comparison matrix, the Raspberry Pi Zero 2 W is the most viable microcontroller for our specific use. Offering efficient, low power consumption along with the capabilities to sustain the PWA encyclopedia and the SQLite Database. It also has integrated Wi-Fi and Bluetooth modules needed to perform proximity handshakes and create a local network. The form factor is very important as well. Its small size enables for a compact and lower cost build. The Pi 4 (4GB) offers the OS we need but uses too much power. The ESP 32 is the most energy efficient being the lowest in power draw both in idle and at full power.

3. High-Endurance Storage

The storage will simply be a High-Endurance MicroSD with a capacity of 64gb to 128gb. Longevity is what we aimed for when choosing a storage device that can handle the write cycles over a massive period of time. We also chose the form factor of an SD card due to “wear-leveling” which writes in all available blocks in order to prevent burning out single memory sectors. This enables it to survive for years or even decades where quality education can finally be met in all countries worldwide.

4. 18650 Power Subsystem

For energy storage, we opted for Four (4) 3400mAh cylindrical lithium-ion cells wired in parallel, creating a massive 13,600mAh reservoir. It's stable, easily sourced, and physically rigid. It can power the microserver for multiple days in harsh climates and extreme weather when solar charging is not an option. This battery bank can supply a steady stream of 3.7V Direct Current (DC) output to the boost converter, discharging slowly while the Pi processes data, and recharges rapidly when the solar panel receives sunlight.

5. Power Regulation

For power regulation, we determined that using a TP4056 with an MT3608 Boost converter is necessary to translate and regulate the electrical current between the solar panel, the battery, and the Pi. The solar panel generates fluctuating voltages depending on the weather and direct sunlight. If we connected the solar panel to the microcontroller directly, it would fry the Pi, damage the internal components, and cease the operations. The TP4056 module receives the fluctuating solar energy and stabilizes the voltage to protect and safely charge the 3.7V 18650 batteries. While that happens, the MT3608 Boost converter takes the 3.7V battery output and pumps it up into a regulated 5V required by the microcontroller. Thus, ensuring a brownout never happens.

6. Off-Grid Energy Harvesting

A single 10W 5V Monocrystalline Solar Panel matches very well with our energy system. It is a rigid photovoltaic panel that transforms sunlight into direct current. It generates up to 2 Amps of current under peak irradiance to charge the 18650 battery bank. We specifically chose a monocrystalline panel because of its highest efficiency rating per square

inch (typically 15%-20%). It is to be mounted to the exterior of a protective enclosure for the LEN, or on the roof of a structure passively harvesting sunlight.

C. Software

The software is an offline local site, which serves as an offline multimedia encyclopedia. It is designed in such a way that the user is exploring a large knowledge base, rather than following a particular learning course. The information is organized as separate hubs for Physics, Mathematics, Chemistry, Geology, Biology, Agriculture, Technology and Computers.

Each major category contains smaller topics. Opening a topic presents an encyclopedia-style article containing text, photographs, diagrams, animations, videos, definitions, examples, and related topics. A search function allows users to quickly find information, while links between related topics encourage users to explore other subjects.

Figure 1. Interface Upon Connection

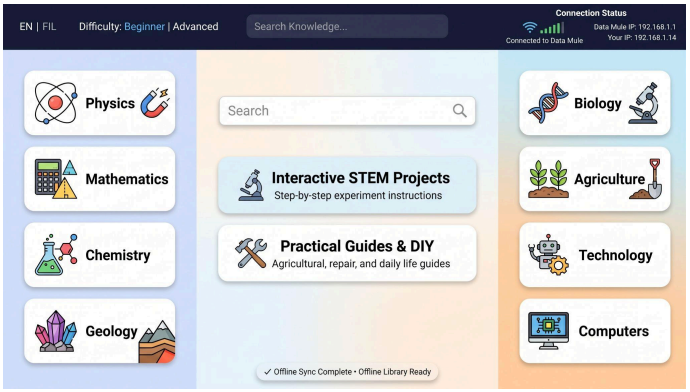


Figure 2. Interface Mobile View



The software is created using standard web technologies like HTML, CSS, and NodeJS. The system uses Nginx as a lightweight local web server to host the encyclopedia. The information is saved in a local SQLite database, whilst the large media files are stored on the same machine as the server. This is to ensure that all resources are local and do not require an internet connection. The NodeJS process

handles requests from attached browsers, SQLite handles the content-related information like articles and topics, and Nginx serves the local site and media files to the user. These technologies allow the ability to host multiple users.

When a user connects their phone, tablet, or computer to the local Wi-Fi network, the browser sends requests to the NodeJS app via the local IP address of the server. The application then pulls necessary text, images, videos, and other data from the local database and storage. A language option will present information in the selected language (Filipino or English). In addition, users' progress, bookmarks, and other data can be saved in local storage and later synced with the Data Mule during the next visit.

The system is designed as a Progressive Web App (PWA) because it is possible for the users to search the content of the online encyclopedia through a browser and have an app-like experience. Therefore, no internet connection is needed **when** the users browse, search, or read the entries.

The software ecosystem will feature a multimedia encyclopedia interface, difficulty level selection, and interactive STEM projects.

Multimedia Encyclopedia Interface

The first feature would be the encyclopedia-like interface. The user would be able to learn new information without being forced into a specific order or structure. Diagrams, photos, animations, and video clips could be integrated into the text of articles. This would allow the user to focus on information that might be dry, difficult, or otherwise inaccessible in a traditional textbook or lecture format.

The system is a Progressive Web Application (PWA) because it is possible for the users to search the content of the offline encyclopedia using their browser but still having them be an app-like experience. Thus, browsing, searching, and reading would not require an Internet connection. For instance, the browser will communicate to the server's local IP address rather than an online website. Upon opening the given address in the browser, the user will be greeted by the encyclopedia's home page, which will host the main subject hubs. Clicking a hub will open its topics, and

clicking a topic will open its articles, related materials, and available STEM projects. The search bar will be able to query the local database, and the language button will toggle the stored content in Filipino and English . The system will also be able to save bookmarks, completed activities, and progress locally, which the Data Mule will sync later.

Difficulty Levels

The same topic can be explained at different levels of detail. Beginner sections would be appropriate for children, whereas advanced sections could include more complex textbook-like explanations and technical terminology. This way children, students, teachers, hobbyists, and the adult population can choose the same system for different purposes

For example, a topic on electricity could start with an overview page about circuits, and include links to more in-depth pages about voltages, currents, resistances, and circuits analysis, letting users go as far as they want to go without forcing them to abandon the program and start over somewhere else.

Interactive STEM Projects

Each feature entry in the textbook/encyclopedia has its own project based on the materials provided in the STEM kits. The project page includes a list of materials, objectives, safety note, step-by-step explanation, diagrams, expected results, and questions. The same system provides simple projects for electricity, magnetism, temperature, light, environmental monitoring, and other subjects where the materials to build are in the learning kit.

Practical Knowledge and Actionable Guides

The encyclopedia is not limited to containing only academic information. It will also have practical guides that can be applied in day-to-day life. The practical guides might include do it yourself (DIY), home improvement, electronics repair, agriculture and gardening, environmental aspects, and other various topics that people might be interested in.

The content provides instructions to the reader. For instance, an agricultural article must not only explain what a given specimen of flora is, but it can also give information on the recommended planting distance, soil, sunlight, watering regimen, and other necessities pertaining to the growth medium of a given plant of interest. Diagrams and illustrations are

provided to further elaborate on relevant measurements, tools, and procedures, thus transforming the encyclopedic resource into a guidebook and a reference material.

Microcontroller Connection

The Local Edge Node (LEN) connects to the Data Mule. The Pi Zero 2 W uses its built-in Wi-Fi to detect and proceed with the proximity handshake with the Data Mule. The handshake allows for pairing and connection of the two devices for the Local Edge Node to receive new content or updates and to send back user progress and other information to the Data Mule. In this way, information can be transferred to and from the system without an internet connection.

Educational Institution Partnerships

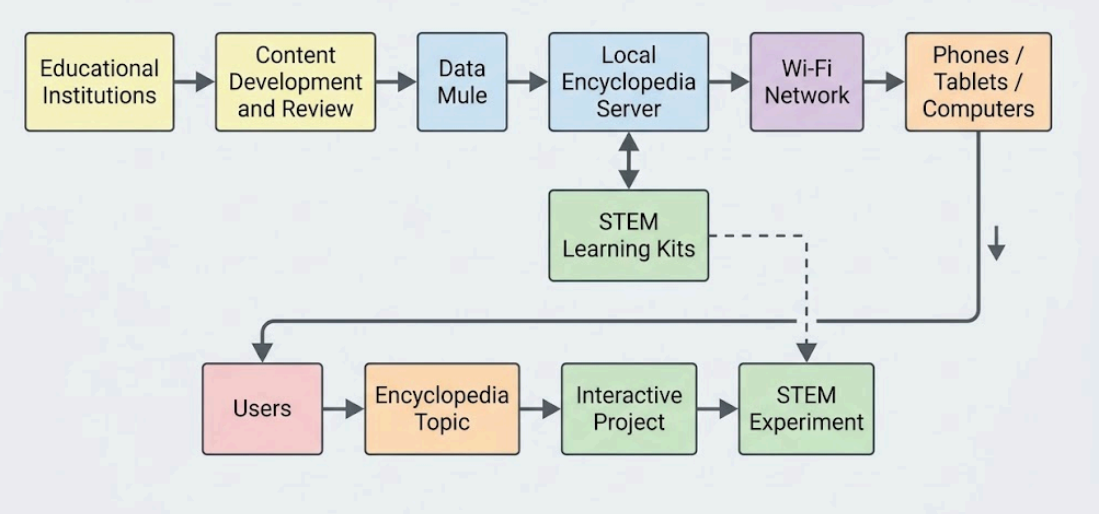
The project will establish partnerships with various academic institutions to provide, develop, review, and update the information provided in the encyclopedia. The academic institutions will not provide generic reading materials. Instead, their faculty members, researchers, and departments can contribute their scientifically reviewed information, experiments, practical guides, and materials in their field of expertise.

System Operation

To begin with, the local server stores the encyclopedia and sets up a Wi-Fi system. Someone connects their phone, tablet, or computer to the network, and then they open the link on their device to access the offline encyclopedia.

Once a topic is chosen, the user may read the article, watch the videos, and explore related topics. If an interactive project is available, the student would then be guided through instructions on how to perform the activity with a STEM kit. They will perform the experiment, make observations, and compare them to the expected results if it is experimental. Projects can also just be instructions to build something that applies the concepts found in the contents.

Figure 3. Conceptual Diagram



III. CHALLENGES/ LIMITATIONS AND MITIGATION STRATEGIES

A. Hardware

Challenges:

Typhoons and extended cloud cover may interrupt solar charging and deplete battery reserves, which compromises power consistency. Heat, humidity, salt-laden air, and flooding may degrade components over time. Storage capacity limits the volume of multimedia content, and the portable unit's Wi-Fi range restricts coverage.

Limitations:

The primary hardware limitation is the system’s dependence on the Data Mule for regular synchronization. If the Data Mule does not reach the Local Edge Node regularly, the delivery of new educational content and the synchronization of student data may be delayed. The amount of data that can be transferred may also be limited by the available data transfer rate and connection conditions, particularly because the proposed system includes large multimedia files such as videos, images, and animations. Another limitation is the Data Mule’s dependence on a reliable power

source. It requires a battery or alternating current (AC) to operate, and continuous use may deplete its battery and require recharging. These hardware limitations may affect the frequency, speed, and amount of information that can be delivered to the Local Edge Node.

Mitigation Strategies:

To address the dependence on alternating current power, the Data Mule is designed to operate primarily from a solar panel connected to a charge controller and a battery bank, with alternating current charging serving only as a supplementary option wherever an electrical outlet exists. A deep-cycle lithium iron phosphate battery is proposed because it tolerates repeated discharge and high temperatures better than ordinary cells. Low-power computing boards, together with automatic sleep states between sessions, reduce energy consumption during continuous use, while a spare battery pack allows operation to continue while the primary battery is recharging. To limit environmental degradation, the electronics are housed in a sealed, ingress-protected enclosure fitted with silicone gaskets and desiccant packs, and the circuit boards receive a protective conformal coating. The unit is mounted above floor level with passive heat sinks and shaded ventilation to manage heat and flood risk. To extend the range of the wireless network, an external high-gain antenna and, where necessary, a second wireless access point near the learning area allow students at a greater distance to connect, and placing the unit centrally within the barangay hall or school further improves coverage. For storage, a high-capacity solid-state drive replaces a basic memory card, and a removable drive bay allows caretakers to install additional or updated drives during visits by partner non-governmental organizations, so that the amount of material is not fixed at the time of deployment.

B. Software

Challenges:

The software must serve a wide range of low-cost devices with no internet connection to fall back on, which makes consistent performance difficult to guarantee.

Because the system operates offline, every correction and every new article depends on someone physically delivering it to the site.

Limitations:

The primary software limitation is that educational content and software updates cannot be applied immediately. New lessons, corrections, and software updates must wait for the next Data Mule synchronization before they can be applied to the Local Edge Node. Compatibility across different operating systems and web browsers may also be a concern, particularly for older or low-cost devices. Some multimedia materials, such as videos, animations, and images, may not function consistently across different devices and browsers. In addition, the system requires regular software maintenance, including bug fixes, content updates, and compatibility testing, to ensure that the platform continues to function properly. These limitations may affect the accessibility, consistency, and reliability of the educational platform.

Mitigation Strategies:

To address compatibility across devices, the software is developed as a Progressive Web Application using standard web technologies, so that it runs inside the browser already installed on iOS and Android phones, tablets, and computers, without requiring separate installation or specialized software. The interface uses a responsive layout and is tested on a range of older, lower-cost handsets before deployment. For multimedia, each video, animation, and image is prepared in several versions, with a widely supported format and reduced resolution as the default, and the server automatically delivers the version best suited to the connecting device. Large files are divided into smaller segments that load progressively, which reduces errors and waiting time on devices with limited memory. Because the system operates without internet connectivity, updates are packaged as verified, numbered bundles that the Data Mule carries to the site and applies automatically at the next visit. The previous version is retained so that the system can be restored if an update fails. Caretakers also receive a simple written guide for basic troubleshooting, such as restarting the server and reconnecting devices, so that minor faults do not require a technician. Finally, a list of tested devices and browsers is documented and shared with partner organizations, so that expectations of performance remain realistic and consistent across communities.

C. Community, Security, and Operational Factors

Challenges:

These challenges arise from the people and conditions surrounding the Base Station. The intended beneficiaries live in low-income and struggling communities, where limited resources, competing livelihood demands, and remote locations affect how consistently the system can be used and maintained.

Limitations:

Because the unit is placed in a shared space, theft, tampering, and unequal access remain risks that no enclosure can fully remove. Repairs are slow in remote barangays, where a technician may be several days away. Households with limited income may be unable to purchase devices, pay for charging, or contribute to maintenance costs, and children may be needed for farming, fishing, or household work, which reduces time for learning. Limited digital literacy and differences between the language of the content and local dialects may further reduce use. Since the Data Mule may be the main source of digital educational resources in the community, many students may connect at the same time, causing slow loading or temporary system failures. Continued operation also depends on external funding, which is vulnerable to changes in donor priorities.

Mitigation Strategies:

The unit is kept in a school or barangay hall under a designated caretaker, in a tamper-evident casing, with orientation provided through partner non-governmental organizations. Modular, replaceable components, a small spare-parts kit, and caretaker training allow minor repairs to be done locally, while system logs carried by the Data Mule support remote diagnosis. The Base Station is free at the point of use, shared tablets and a charging point reduce household device costs, and sessions scheduled outside working hours accommodate livelihood demands. Content is offered in Filipino and, where feasible, major regional languages. A limit on simultaneous connections, together with stored copies of frequently opened pages, protects performance under heavy use. Partnerships with several non-governmental organizations and local government units reduce reliance on a single funding source.

IV. CONCLUSION

This concept paper proposed the Local Edge Node and Data Mule system as a means of delivering digital educational resources to off-grid communities in the Philippines, where the absence of electricity and affordable internet access continues to separate learners from modern educational technologies. A solar-powered, low-power micro-server hosts an offline multimedia encyclopedia and serves learners through a local wireless network, while a Data Mule carries updated content and student progress between the community and its urban partners. Supported by Science, Technology, Engineering, and Mathematics learning kits and contributions from partner academic institutions, the system advances Sustainable Development Goal 4 on inclusive and equitable quality education. The paper also acknowledges that technical design alone does not guarantee success. Power interruptions during typhoons, environmental wear, limited storage and range, device compatibility, delayed updates, theft, slow repairs, household poverty, limited digital literacy, and dependence on external funding may each reduce effectiveness. Mitigation strategies were therefore proposed at the hardware, software, and community levels, including hybrid power, protective enclosures, adaptive media, verified update bundles, trained caretakers, and partnerships with non-governmental organizations and local government units. Because this work remains a concept, field testing in off-grid barangays is recommended to validate performance, durability, and learning outcomes before wider deployment

V. AI Disclosure

AI image generation has been used to create the diagrams (Figure 1, Figure 2, and Figure 3). AI writing assistants have also been used to assist in writing, correcting, and editing the text.

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